English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, 12 June 2022 Test 2 — *Your grade is 2 + the number of points. Time allowed: 3 hours. Good luck! Throughout, work on* $(\Omega, \mathcal{F}, \mathbb{P})$*, and let* X, Y *denote random variables.*

Problem 1. Let $f_{X,Y}(x,y) = cx + 1$ for $x, y \geq 0$, $x + y \leq 1$, and 0 otherwise. Find:

1. (0.75 points) c and $\operatorname{Cov}(X, Y)$;
2. (0.25 points) $\mathbb{E}[X \mid Y = 1/2]$.

Problem 2. The weight of peas in a type of tin is modelled by $N(\mu, 10^2)$ grams. Fifteen percent of tins contain less than 250 grams. Find:

1. (0.25 points) μ;
2. (0.25 points) the percentage containing more than 280 grams.

After a change, the model is $N(250, \sigma^2)$.

3. (0.5 points) Find σ if 97% of tins contain between 230 and 270 grams.

Problem 3. A Probability and Statistics test has four independent problems, solved with probabilities 80%, 70%, 60% and 40%. Each awards 10 points if solved and 0 otherwise; the grade is 2 + points/10. For 100 students, calculate:

1. (0.25 points) the probability that the average score on problem 1 is below 8;
2. (0.25 points) the expected average grade;
3. (0.5 points) the probability that the average grade is at least 1% above its expectation.

Problem 4. Insurer "Ins 1" models claims as independent exponential variables with mean 100 BGN.[1] It buys reinsurance for $x > 0$ BGN from "Ins 2": if a claim exceeds 300 BGN, Ins 2 pays 200 BGN of it.

1. (0.25 points) What is the probability that Ins 1 pays less than 200 BGN from its own funds for one claim?
2. (0.25 points) If Ins 2 wants mean profit 10 BGN per claim, what should x be?
3. (0.5 points) If Ins 1 wants mean profit 10 BGN per claim, what should its policy premium be? What would it be without reinsurance? Why buy reinsurance if the mean profit is unchanged?

[1] Recall that $X \sim \operatorname{Exp}(\lambda)$ has density $\lambda e^{-\lambda x} \mathbb{1}_{\{x>0\}}$.